CS 70 Discrete Mathematics and Probability Theory
Spring 2026 Course Notes **Note 1**

1 Propositional Logic

In order to be fluent in working with mathematical statements, you need to understand the basic framework of the language of mathematics. In this first lecture, we will start by learning about what logical forms mathematical theorems may take, and how to manipulate those forms to make them easier to prove. In the next few lectures, we will learn several different methods of proving things.

Our first building block is the notion of a **proposition**, which is simply a statement which is either true or false.
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These statements are all propositions:

(1) $\sqrt{3}$ is irrational.

(2) $1+1=5$.

(3) Julius Caesar had 2 eggs for breakfast on his 10th birthday.

These statements are clearly not propositions:

(4) $2+2$.

(5) $x^2+3x=5$. [What is x?]

These statements aren't propositions either (although some books say they are). Propositions should not include fuzzy terms.

(6) Arnold Schwarzenegger often eats broccoli. [What is "often?"]

(7) Henry VIII was unpopular. [What is "unpopular?"]

Propositions may be joined together to form more complex statements. Let P and Q be variables representing propositions (for example, P could stand for "3 is odd"). The simplest way of joining these propositions together is to use the connectives "and", "or" and "not."

(1) **Conjunction**: $P \wedge Q$ ("P and Q"). True only when both P and Q are true. \land

(2) **Disjunction**: $P \vee Q$ ("P or Q"). True when at least one of P and Q is true. \|or

(3) **Negation**: $\neg P$ ("not P"). True when P is false. \neg

Statements like these, with variables, are called *propositional forms*.

A fundamental principle known as the **law of the excluded middle** 排中律 says that, for any proposition P, either P is true or $\neg P$ is true (but not both). Thus $P \vee \neg P$ is always true, regardless of the truth value of P. A propositional form that is always true regardless of the truth values of its variables is called a *tautology*. Conversely, a statement such as $P \wedge \neg P$, which is always false, is called a *contradiction*.

Concept check! If we let P stand for the proposition "3 is odd", Q stand for "4 is odd", and R for "$4+5=49$", what are the values of $P \wedge R$, $P \vee R$ and $\neg Q$?

false true true

A useful tool for describing the possible values of a propositional form is a **truth table**. Truth tables are the same as function tables: you list all possible input values for the variables, and then list the outputs given those inputs. (The order does not matter.)

Here are truth tables for conjunction and negation:

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

P	$\neg P$
T	F
F	T

Concept check! Write down the truth table for disjunction (OR).

The most important and subtle propositional form is an **implication**:

(4) **Implication**: $P \implies Q$ ("P implies Q"). This is the same as "If P, then Q."

Here, P is called the *hypothesis* of the implication, and Q is the *conclusion*.[1]

We encounter implications frequently in everyday life; here are a couple of examples:

> If you stand in the rain, then you'll get wet.
>
> If you passed the class, you received a certificate.

An implication $P \implies Q$ is false only when P is true and Q is false. For example, the first statement above would be false only if you stood in the rain but didn't get wet. The second statement would be false only if you passed the class but didn't receive a certificate.

Here is the truth table for $P \implies Q$ (along with an extra column that we'll explain in a moment):

P	Q	$P \implies Q$	$\neg P \vee Q$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

Note that $P \implies Q$ is always true when P is false. This means that many statements that sound nonsensical in English are true, mathematically speaking. Examples are statements like: "If pigs can fly, then horses can read" or "If 14 is odd then $1+2=18$." (These are statements that we never make in everyday life, but are perfectly natural in mathematics.) When an implication is stupidly true because the hypothesis is false, we say that it is *vacuously true*.

Note also that $P \implies Q$ is **logically equivalent** to $\neg P \vee Q$, as can be seen by comparing the last two columns of the above truth table: for all possible truth values of P and Q, $P \implies Q$ and $\neg P \vee Q$ take the same values

[1] P is also sometimes called the *antecedent* and Q the *consequent*.

(i.e., they have the same truth table). We write this as $(P \implies Q) \equiv (\neg P \vee Q)$.

$P \implies Q$ is the most common form mathematical theorems take. Here are some of the different ways of saying it:

(1) if P, then Q;

(2) Q if P;

(3) P only if Q;

(4) P is sufficient for Q;

(5) Q is necessary for P;

(6) Q unless not P.

If both $P \implies Q$ and $Q \implies P$ are true, then we say "P if and only if Q" (abbreviated "P iff Q"). Formally, we write $P \iff Q$. Note that $P \iff Q$ is true only when P and Q have the same truth values (both true or both false).

For example, if we let P be "3 is odd," Q be "4 is odd," and R be "6 is even," then the following are all true: $P \implies R$, $Q \implies P$ (vacuously), and $R \implies P$. Because $P \implies R$ and $R \implies P$, we also see that $P \iff R$ is true.

Given an implication $P \implies Q$, we can also define its

(a) **Contrapositive**: $\neg Q \implies \neg P$

(b) **Converse**: $Q \implies P$

The contrapositive of "If you passed the class, you received a certificate" is "If you did not get a certificate, you did not pass the class." The converse is "If you got a certificate, you passed the class." Does the contrapositive say the same thing as the original statement? Does the converse?

Let's look at the truth tables:

P	Q	$\neg P$	$\neg Q$	$P \implies Q$	$Q \implies P$	$\neg Q \implies \neg P$	$P \iff Q$
T	T	F	F	T	T	T	T
T	F	F	T	F	T	F	F
F	T	T	F	T	F	T	F
F	F	T	T	T	T	T	T

Note that $P \implies Q$ and its contrapositive have the *same* truth values everywhere in their truth tables, so they are logically equivalent: $(P \implies Q) \equiv (\neg Q \implies \neg P)$. Many students confuse the contrapositive with the converse: note that $P \implies Q$ and $\neg Q \implies \neg P$ are logically equivalent, but $P \implies Q$ and $Q \implies P$ are not!

When two propositional forms are logically equivalent, we can think of them as "meaning the same thing." We will see in the next lecture how useful this can be for proving theorems.

2 Quantifiers

The mathematical statements you'll see in practice will not be made up of simple propositions like "3 is odd." Instead you'll see statements like:

(1) For all natural numbers n, $n^2 + n + 41$ is prime.

(2) If n is an odd integer, so is n^3.

(3) There is an integer k that is both even and odd.

In essence, these statements assert something about lots of simple propositions (even infinitely many!) all at once. For instance, the first statement is asserting that 0^2+0+41 is prime, 1^2+1+41 is prime, and so on. The last statement is saying that, as k ranges over all possible integers, we will find at least one value of k for which the statement is satisfied.

Why are the above three examples considered to be propositions, while earlier we claimed that "$x^2+3x=5$" was not? The reason is that in these examples, there is an underlying "universe" that we are working in, and the statements are *quantified* over that universe. To express these statements mathematically we need two **quantifiers**: The *universal quantifier* $\forall$ ("for all") and the *existential quantifier* $\exists$ ("there exists"). Examples:

(1) "Some mammals lay eggs." Mathematically, "some" means "at least one," so the statement is saying "There exists a mammal x such that x lays eggs." If we let our universe U be the set of mammals, then we can write: $(\exists x \in U)(x \text{ lays eggs})$. (Sometimes, when the universe is clear, we omit U and simply write $\exists x(x \text{ lays eggs})$.)

(2) "For all natural numbers n, n^2+n+41 is prime," can be expressed by taking our universe to be the set of natural numbers, denoted as $\mathbb{N}$: $(\forall n \in \mathbb{N})(n^2+n+41 \text{ is prime})$.

We refer to a statement which refers to a variable as a *predicate* or as a *propositional formula* when replacing the variable with a value makes the statement either true or false. For example, the statement "n^2+n+41 is prime" is a predicate or propositional formula with variable n. The value of the statement on natural numbers, n, is either true or false depending on the value of n. That is, replacing the variable with a value makes the predicate into a proposition. In particular, for $n=1$, we have "43 is prime" which is true. Where for $n=41$, the statement "$41^2+41+41$ is prime" is false as one can factor $41^2+41+41$ into 41×43.

Note that in a *finite* universe, we can express existentially and universally quantified propositions without quantifiers, using disjunctions and conjunctions respectively. For example, if our universe U is $\{1,2,3,4\}$, then $(\exists x \in U)P(x)$ is logically equivalent to $P(1)\vee P(2)\vee P(3)\vee P(4)$, and $(\forall x \in U)P(x)$ is logically equivalent to $P(1)\wedge P(2)\wedge P(3)\wedge P(4)$. However, in an infinite universe, such as the natural numbers, this is not possible.

Concept check! Use quantifiers to express the following two statements: "For all integers x, $2x+1$ is odd", and "There exists an integer between 2 and 4".

Some statements can have multiple quantifiers. As we will see, however, quantifiers do not commute. You can see this just by thinking about English statements. Consider the following (rather gory) example:

> "Every time I ride the subway in New York, someone gets stabbed."
>
> "There is someone, such that every time I ride the subway in New York, that someone gets stabbed."

The first statement is saying that every time I ride the subway someone gets stabbed, but it could be a different person each time. The second statement is saying something truly horrible: that there is some poor guy Joe with the misfortune that every time I get on the New York subway, there is Joe, getting stabbed again. (Poor Joe will run for his life the second he sees me.)

Mathematically, we are quantifying over two universes: $T=\{\text{times when I ride on the subway}\}$ and $P=\{\text{people}\}$. The first statement can be written: $(\forall t \in T)(\exists p \in P)(p \text{ gets stabbed at time } t)$. The second

statement says: $(\exists p \in P)(\forall t \in T)(p$ gets stabbed at time $t)$.

Let's look at a more mathematical example:

Consider

1. $(\forall x \in \mathbb{Z})(\exists y \in \mathbb{Z})(x < y)$
2. $(\exists y \in \mathbb{Z})(\forall x \in \mathbb{Z})(x < y)$

The first statement says that, given an integer, I can find a larger one. The second statement says something very different: that there is a largest integer! The first statement is true, the second is not.

3 Much Ado About Negation

What does it mean for a proposition P to be false? It means that its negation $\neg P$ is true. It's helpful to have some rules for working with negation, as will become more obvious next lecture when we look at proofs.

First, let's look at how to negate conjunctions and disjunctions:

$$\neg(P \wedge Q) \equiv (\neg P \vee \neg Q)$$

$$\neg(P \vee Q) \equiv (\neg P \wedge \neg Q)$$

These two equivalences are known as *De Morgan's Laws*, and they are quite intuitive: for example, if it is not the case that $P \wedge Q$ is true, then either P or Q must be false (and vice versa).

Concept check! Verify both of De Morgan's Laws by writing down the appropriate truth tables.

Negating propositions involving quantifiers actually follows analogous laws. Let's start with a simple example. Assume that the universe is $\{1,2,3,4\}$ and let $P(x)$ denote the propositional formula "$x^2 > 10$." Check that $\exists x P(x)$ is true but $\forall x P(x)$ is false. Observe that both $\neg(\forall x P(x))$ and $\exists x \neg P(x)$ are true because $P(1)$ is false. Also note that both $\forall x \neg P(x)$ and $\neg(\exists x P(x))$ are false, since $P(4)$ is true. The fact that each pair of statements had the same truth value is no accident, as the equivalences

$$\neg(\forall x P(x)) \equiv \exists x \neg P(x)$$

$$\neg(\exists x P(x)) \equiv \forall x \neg P(x)$$

are laws that hold for any proposition P quantified over any universe (including infinite ones).

It is helpful to think of English sentences to convince yourself (informally) that these laws are true. For example, assume that we are working within the universe $\mathbb{Z}$ (the set of all integers), and that $P(x)$ is the proposition "x is odd." We know that the statement $(\forall x P(x))$ is false, since not every integer is odd. Therefore, we expect its negation, $\neg(\forall x P(x))$, to be true. But how would you say the negation in English? Well, if it is not true that every integer is odd, then there must exist some integer which is not odd (i.e., even). How would this be written in propositional form? That's easy, it's just: $(\exists x \neg P(x))$.

To see a more complex example, fix some universe and propositional formula $P(x,y)$. Assume we have the proposition $\neg(\forall x \exists y P(x,y))$ and we want to push the negation operator inside the quantifiers. By the above laws, we can do it as follows:

$$\neg(\forall x \exists y P(x,y)) \equiv \exists x \neg(\exists y P(x,y)) \equiv \exists x \forall y \neg P(x,y).$$

Notice that we broke the complex negation into a smaller, easier problem as the negation propagated itself through the quantifiers. Note also that the quantifiers "flip" as we go.

4 Trickier Quantifier Examples

Let's look at a trickier set of examples:

Write the sentence "there are **at least** three distinct integers x that satisfy $P(x)$" as a proposition using quantifiers! One way to do it is

$$\exists x \exists y \exists z (x \neq y \wedge y \neq z \wedge z \neq x \wedge P(x) \wedge P(y) \wedge P(z)).$$

(Here all quantifiers are over the universe $\mathbb{Z}$ of integers.)

Now write the sentence "there are **at most** three distinct integers x that satisfy $P(x)$" as a proposition using quantifiers. One way to do it is

$$\exists x \exists y \exists z \forall d (P(d) \implies d = x \vee d = y \vee d = z).$$

If you're not sure how to interpret this propositional logic statement, you can read it as: "There exist 3 three specific integers x, y, and z, such that if $P(d)$ is true, then d has to be one of those three integers." There are many other ways to express this same statement, for example:

$$\forall x \forall y \forall v \forall z ((x \neq y \wedge y \neq v \wedge v \neq x \wedge x \neq z \wedge y \neq z \wedge v \neq z) \implies \neg(P(x) \wedge P(y) \wedge P(v) \wedge P(z))).$$

Concept check! Check that you understand both of the above alternatives.

Note that we can also generate even more exactly identical alternatives, e.g. writing the contrapositives, propagating negations, etc. In general, you want to pick the propositional logic statement that most clearly expresses the idea.

Finally, what if we want to express the sentence "there are **exactly** three distinct integers x that satisfy $P(x)$"? This is now easy: we can just use the *conjunction* of the two propositions above.

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∧ conjuction \\and 恒真 重言 tautology

∨ disjuction \\or contradiction

logic